Jacobi Matching Matrices: Endpoint Singularities, Gamma-Function Arithmetic, and Bernstein Zeros

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Abstract

For a weight W on a compact interval with pure endpoint singularities we study the *matching matrix* $b_{mk} = \int \widehat{P}_m \widehat{P}_k W^{3/2}$, where $\widehat{P}_n$ are the orthonormal polynomials for W. Taking $W_{\alpha\beta} = (1-x)^\alpha (1+x)^\beta$ on $[-1,1]$ (Jacobi weight), we treat four cases: Chebyshev-T ($\alpha = \beta = -\frac{1}{2}$), Chebyshev-U ($\alpha = \beta = \frac{1}{2}$), Legendre ($\alpha = \beta = 0$), an asymmetric case ($\alpha = \frac{1}{2}, \beta = -\frac{1}{2}$), and a sextic test case ($\alpha = \beta = -\frac{5}{9}$).

Using the substitution $x = \cos\theta$ and the classical Wallis-type formula

$$\int_0^\pi \sin^p\theta \cos(q\theta)\, d\theta = \cos\!\left(\tfrac{\pi q}{2}\right) \frac{\pi\, \Gamma(p{+}1)}{2^p\, \Gamma\!\left(\frac{p+q+2}{2}\right) \Gamma\!\left(\frac{p-q+2}{2}\right)},$$

we obtain *exact closed forms* for the entire Chebyshev matching tables. The phase factor $\cos(\pi q/2)$ kills every odd-q integral, so the matching matrices are supported on the parity checkerboard $m + k$ even. The main results are:

1. **Decay theorem.** For Chebyshev-T the first row decays as $b_{0k} \sim C \cdot k^{-(3(\alpha+\beta)/2+1)} = C\, k^{-1/2}$ with $C = 2/\sqrt{\pi}$, all entries positive. For Chebyshev-U the two Wallis terms cancel at leading order and the true decay is $k^{-7/2}$, one power *faster* than the naive law $k^{-5/2}$.
2. **Generator ring.** All Chebyshev matching entries (T and U) lie in $\mathbb{Q}(\sqrt{2}) \cdot \gamma_{\mathrm{e}}$, where $\gamma_{\mathrm{e}} = \Gamma(\frac{1}{4})^2\sqrt{2}/(2\pi^{3/2})$ is a *single* transcendental generator (the $\sqrt{2}$ enters only through the normalisation ν_0/ν_n).
3. **Legendre fixed point.** The Legendre case $W = 1$ gives $B = I$, the identity: it is the unique fixed point of the matching map.
4. **Sextic test and connection–Gram theorem.** For general symmetric α the matching matrix is the Gram matrix of the Gegenbauer connection coefficients $C^{(\lambda)} \to C^{(\mu)}$, $\lambda = \alpha + \frac{1}{2}$, $\mu = \frac{3}{2}\alpha + \frac{1}{2}$. At $\alpha = -\frac{5}{9}$ the generator is the sextic period $B(\frac{1}{6}, \frac{1}{2})$ (the curve $y^2 = 1 - x^6$, CM by $\mathbb{Z}[\omega]$), with moment ratios stepping in thirds and polynomial data in eighteenths; the Chebyshev closed forms are the rational specialisation $\alpha = -\frac{1}{2}$.
5. **Bernstein zeros.** Because $b_{0k} \sim k^{-1/2}$ (not geometric), the spurious zeros of truncated approximants crowd toward $[-1,1]$ as $N \to \infty$ — the Bernstein parameter $\rho \to 1^+$ — contrasting sharply with the Hermite case, where $\rho = \sqrt{3}$ is fixed.

This family completes the *weight triptych*: entire (Hermite), pole at finite distance (Meixner–Pollaczek), and pure endpoint singularity (Jacobi).

1 Setup and Motivation

1.1 The matching matrix

Let $W\colon X \to \mathbb{R}_{\geq 0}$ be a weight, and let $(\widehat{P}_n)_{n\geq 0}$ be the associated orthonormal polynomials: $\int \widehat{P}_m \widehat{P}_k\, W\, dx = \delta_{mk}$. Define the *Jacobi functions* $\Phi_n = \widehat{P}_n \sqrt{W}$ (orthonormal in $L^2(dx)$). The **matching matrix** is

$$b_{mk} \;=\; \langle \Phi_m, \Phi_k \rangle_W \;=\; \int \widehat{P}_m\, \widehat{P}_k\, W^{3/2}\, dx,$$

the overlap of the L^2-orthonormal functions against the weight W itself. When W is normalised ($\int W\,dx = 1$), $b_{00} = 1$ and the first row $(b_{0k})_{k\geq 0}$ is the expansion of $\sqrt{W}$ in the basis $(\widehat{P}_k\sqrt{W})$.

1.2 Calibration: the Hermite and MP cases

Prior work [1, 2] established:

- **Hermite** ($W = e^{-x^2}$, entire): $b_{mk} \in \mathbb{Z}$ after normalisation; geometric decay $b_{0k} \sim 3^{-k/2}$; spurious zeros of the approximants $A_N = \sum_{k\leq N} b_{0k}\widehat{P}_k$ accumulate on a fixed ellipse ("Bernstein ellipse" $\rho = \sqrt{3}$).
- **Meixner–Pollaczek** ($\lambda = \frac{1}{4}$, poles at finite distance): power-law decay $b_{0k} \sim k^{-\alpha}$ with $\alpha \approx \frac{3}{4}$; spurious zeros accumulate near the next Bump–Ng line.

The present paper handles the *third vertex* of the triptych: compact support with *pure endpoint singularities*.

2 Jacobi Weights and the Wallis Formula

2.1 Four cases

We study $W_{\alpha\beta}(x) = (1-x)^\alpha(1+x)^\beta$ on $[-1, 1]$ in four cases:

Case	Parameters	Character
Chebyshev-T	$\alpha = \beta = -\frac{1}{2}$	Endpoint singularity; $W^{3/2} = (1-x^2)^{-3/4}$
Chebyshev-U	$\alpha = \beta = +\frac{1}{2}$	Endpoint vanishing; $W^{3/2} = (1-x^2)^{+3/4}$
Legendre	$\alpha = \beta = 0$	Flat weight; $W = 1$
Asymmetric	$\alpha = \frac{1}{2}, \beta = -\frac{1}{2}$	One endpoint singular, one vanishing

2.2 Substitution $x = \cos\theta$ and the Wallis integral

Under $x = \cos\theta$, $dx = -\sin\theta\,d\theta$, and $(1-x^2)^s = \sin^{2s}\theta$, so

$$\int_{-1}^{1} f(x)\,(1-x^2)^s\,dx = \int_0^\pi f(\cos\theta)\,\sin^{2s+1}\theta\,d\theta.$$

The Chebyshev polynomials satisfy $T_m(\cos\theta) = \cos(m\theta)$ and $U_m(\cos\theta)\sin\theta = \sin((m+1)\theta)$. The key integral is the **Wallis-type formula**: for $p > -1$ and $q \geq 0$ integer,

$$\mathcal{I}(p,q) \;:=\; \int_0^\pi \sin^p\theta\cos(q\theta)\,d\theta \;=\; \cos\Big(\frac{\pi q}{2}\Big)\,\frac{\pi\,\Gamma(p+1)}{2^p\,\Gamma\Big(\frac{p+q+2}{2}\Big)\,\Gamma\Big(\frac{p-q+2}{2}\Big)}. \tag{1}$$

This follows from $\int_0^\pi e^{iq\theta}\sin^p\theta\,d\theta = \pi e^{i\pi q/2}\Gamma(p+1)/\big(2^p\,\Gamma(\frac{p+q+2}{2})\Gamma(\frac{p-q+2}{2})\big)$ (a Beta-function evaluation) by taking real parts. **The phase factor is essential**: $\cos(\pi q/2)$ vanishes for all odd q and equals $(-1)^{q/2}$ for even q; the latter sign exactly cancels the sign of $\Gamma(\frac{p-q+2}{2})$ (which alternates through the poles of Γ), so for $p = -\frac{1}{2}$ every even-q integral is *positive*. Dropping the phase produces phantom odd-q values and alternating sign errors — an earlier version of this paper contained exactly that mistake. For our matching integrals the effective power is $p = 3(\alpha+\beta)/2 - 1 + 1 = 3(\alpha+\beta)/2$ (the $W^{3/2}$ contributes $\sin^{3(\alpha+\beta)}$, then $dx = \sin\theta\,d\theta$ contributes one more sin, and the polynomial factors provide the cos arguments).

3 Chebyshev-T: Exact Closed Forms

3.1 Orthonormal polynomials and matching integral

The orthonormal Chebyshev-T polynomials are $\widehat{P}_0^T = 1/\sqrt{\pi}$ and $\widehat{P}_n^T = \sqrt{2/\pi}\,T_n$ for $n \geq 1$. The matching matrix is

$$b_{mk}^T = \int_{-1}^{1} \widehat{P}_m^T(x)\,\widehat{P}_k^T(x)\,(1-x^2)^{-3/4}\,dx.$$

Under $x = \cos\theta$ the integrand becomes $\cos(m\theta)\cos(k\theta)\sin^{-1/2}\theta$ (after the dx substitution), so

$$b_{mk}^T = \nu_m \nu_k \cdot \frac{\mathcal{I}(-\frac{1}{2}, m+k) + \mathcal{I}(-\frac{1}{2}, |m-k|)}{2},$$

where $\nu_0 = 1/\sqrt{\pi}$ and $\nu_n = \sqrt{2/\pi}$ for $n \geq 1$.

3.2 Closed form via Gamma functions

Using (1) and the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$:

Proposition 3.1. *Let* $\gamma_{\mathrm{e}} = \Gamma(\frac{1}{4})^2\sqrt{2}/(2\pi^{3/2}) = \mathcal{I}(-\frac{1}{2}, 0)/\pi$, *and write*

$$r(n) = \frac{\mathcal{I}(-\frac{1}{2}, 2n)}{\mathcal{I}(-\frac{1}{2}, 0)} = \frac{(\frac{1}{4})_n}{(\frac{3}{4})_n} = \frac{1\cdot 5\cdots(4n-3)}{3\cdot 7\cdots(4n-1)} \;>\; 0.$$

Then $b_{mk}^T = 0$ *whenever* $m+k$ *is odd, and for* $m+k$ *even (with* $s = \frac{m+k}{2}$, $d = \frac{|m-k|}{2}$*):*

$$b_{mk}^T \;=\; \begin{cases} \gamma_{\mathrm{e}} & m = k = 0, \\ \sqrt{2}\,\gamma_{\mathrm{e}}\, r(s) & m = 0,\ k \geq 2 \text{ even}, \\ \gamma_{\mathrm{e}}\,\big[r(d) + r(s)\big] & m, k \geq 1. \end{cases}$$

Every entry is a positive rational multiple of γ_{e} *or* $\sqrt{2}\,\gamma_{\mathrm{e}}$*: all entries lie in* $\mathbb{Q}_{>0}(\sqrt{2})\cdot\gamma_{\mathrm{e}}$*, a* ***single transcendental generator*** *over* $\mathbb{Q}(\sqrt{2})$*, with the* $\sqrt{2}$ *a pure normalisation artefact (*$\nu_0 = \nu_n/\sqrt{2}$*).*

3.3 Exact 5×5 table

Table 1: Chebyshev-T matching matrix b_{mk}^T (upper triangle, $0 \leq m \leq k \leq 4$). All entries exact and positive; $\gamma_{\mathrm{e}} \approx 1.6693$. Entries with $m+k$ odd vanish identically. Verified by `mpmath` quadrature of the defining integral at 40 digits.

$m\backslash k$	0	1	2	3	4
0	$1\cdot\gamma_{\mathrm{e}}$	0	$\frac{1}{3}\sqrt{2}\,\gamma_{\mathrm{e}}$	0	$\frac{5}{21}\sqrt{2}\,\gamma_{\mathrm{e}}$
1		$\frac{4}{3}\gamma_{\mathrm{e}}$	0	$\frac{4}{7}\gamma_{\mathrm{e}}$	0
2			$\frac{26}{21}\gamma_{\mathrm{e}}$	0	$\frac{122}{231}\gamma_{\mathrm{e}}$
3				$\frac{92}{77}\gamma_{\mathrm{e}}$	0
4					$\frac{90}{77}\gamma_{\mathrm{e}}$

The vanishing of the odd-parity entries is forced by the symmetry $\theta \mapsto \pi - \theta$ of $\sin^{-1/2}\theta$: only even frequencies survive. The diagonal rationals $b_{nn}^T/\gamma_{\mathrm{e}} = 1 + r(n)$ are $1,\ \frac{4}{3},\ \frac{26}{21},\ \frac{92}{77},\ \frac{90}{77},\ \frac{1684}{1463}, \ldots$, which for $n \geq 1$ decrease monotonically to 1 as $r(n) \sim c\,n^{-1/2} \to 0$.

4 Decay Theorem

Theorem 4.1 (Endpoint-singularity decay law)**.** *Along even k (odd entries vanish identically):*

$$b_{0k}^T \;=\; \sqrt{\tfrac{2}{\pi}}\,\frac{\Gamma(\frac{k}{2}+\frac{1}{4})}{\Gamma(\frac{k}{2}+\frac{3}{4})} \;=\; \sqrt{2}\,\gamma_{\mathrm{e}}\,r(\tfrac{k}{2}) \;\sim\; \frac{2}{\sqrt{\pi}}\,k^{-1/2}, \qquad b_{0k}^U \;\sim\; -\,C_U\,k^{-7/2},$$

with all $b_{0k}^T > 0$ and all $b_{0k}^U < 0$ for $k \geq 2$ even. The Chebyshev-T exponent agrees with the naive endpoint law $k^{-(3(\alpha+\beta)/2+1)}$; the Chebyshev-U exponent does not: the naive law predicts $k^{-5/2}$, but the two Wallis terms $\mathcal{I}(\frac{3}{2},k) - \mathcal{I}(\frac{3}{2},k{+}2)$ cancel at leading order, and the exact telescoping ratio $\mathcal{I}(\frac{3}{2},2n) - \mathcal{I}(\frac{3}{2},2n{+}2) = \mathcal{I}(\frac{3}{2},2n)\cdot\frac{5/2}{n+7/4}$ costs one extra power of k.

Concretely:

Case	$\alpha+\beta$	*Exponent*	*Asymptotic*
Chebyshev-T	-1	$-1/2$	$b_{0k}^T \sim C_T/\sqrt{k},\ C_T = 2/\sqrt{\pi}$
Chebyshev-U	$+1$	$-7/2$	$b_{0k}^U \sim -C_U/k^{7/2},\ C_U = \frac{15}{2\sqrt{2\pi}}$
Legendre	0	*degenerate*	$b_{0k}^L = \delta_{0k}$

Proof sketch. Apply the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ to $\Gamma(\frac{p-q+2}{2})$ in (1) with $q = 2n$ even. For $p = -\frac{1}{2}$ this gives the exact evaluation $\mathcal{I}(-\frac{1}{2},2n) = \sqrt{\pi}\,\Gamma(n+\frac{1}{4})/\Gamma(n+\frac{3}{4}) > 0$ (the phase $(-1)^n$ cancels against the sign of the reflected Gamma), and Stirling gives $\Gamma(n+\frac{1}{4})/\Gamma(n+\frac{3}{4}) \sim n^{-1/2}$, hence $b_{0k}^T \sim 2/\sqrt{\pi k}$. For $p = \frac{3}{2}$ one finds $\mathcal{I}(\frac{3}{2},2n) = -\frac{1}{4}\Gamma(\frac{5}{2})\,\Gamma(n-\frac{3}{4})/\Gamma(n+\frac{7}{4}) < 0$ for $n \geq 1$, of order $n^{-5/2}$; but $b_{0k}^U \propto \mathcal{I}(\frac{3}{2},k) - \mathcal{I}(\frac{3}{2},k{+}2)$ and the ratio of consecutive terms is $\Gamma(n-\frac{3}{4})/\Gamma(n+\frac{7}{4}) \mapsto$ multiply by $(n-\frac{3}{4})/(n+\frac{7}{4})$, so the difference is $\mathcal{I}(\frac{3}{2},2n)\cdot\frac{5/2}{n+7/4} = O(n^{-7/2})$. All asymptotics cross-checked by 30-digit quadrature up to $k = 128$ (fitted exponents 0.500 and $\to 3.5$). □ □

The key *mechanism*: the endpoint exponent determines the effective power p of $\sin\theta$ in the angular integral, and Stirling converts p into the decay exponent $-(p+1)$ — *provided a single Wallis term dominates.* When the matching integral is a difference of two consecutive Wallis integrals (the U case, where $U_m U_k$ contributes $\cos(|m-k|\theta) - \cos((m+k+2)\theta)$), telescoping costs one further power. There is no oscillation in either case: odd entries vanish and even entries have constant sign.

5 Legendre: The Fixed Point

Proposition 5.1 (Legendre is a fixed point)**.** *For $\alpha = \beta = 0$, $W \equiv 1$ on $[-1,1]$. Then $W^{3/2} = W = 1$ and $b_{mk}^L = \int_{-1}^{1} \widehat{P}_m^L\,\widehat{P}_k^L\,dx = \delta_{mk}$. The matching matrix is the identity: $B^L = I$.*

This is the **unique fixed point** of the matching map $W \mapsto W^{3/2}$ among Jacobi weights: only the flat weight $W \equiv 1$ satisfies $W^{3/2} \propto W$. (Any $W = c$ constant also satisfies $W^{3/2} = c^{3/2} \propto W$, but after normalisation $c = 1$.)

To have a nontrivial fourth data point we study the asymmetric case $\alpha = \frac{1}{2}, \beta = -\frac{1}{2}$ in §6: its first row is exactly rational after dividing by $\sqrt{2}$, alternates in sign, and decays as $k^{-1/2}$.

6 Chebyshev-U and Asymmetric Cases

6.1 Chebyshev-U closed form

Using $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$, the matching integrand becomes

$$U_m(\cos\theta)\,U_k(\cos\theta)\,(1-\cos^2\theta)^{3/4}\sin\theta = \frac{\sin((m+1)\theta)\sin((k+1)\theta)}{\sin\theta}\cdot\sin^{3/2}\theta\cdot\sin\theta,$$

and the product-to-sum formula gives

$$b^U_{mk} = \frac{1}{\pi/2} \cdot \frac{\mathcal{I}(\frac{3}{2}, |m-k|) - \mathcal{I}(\frac{3}{2}, m+k+2)}{2}.$$

Both arguments have the parity of $m+k$, so $b^U_{mk} = 0$ for $m+k$ odd, exactly as in the T case. For even arguments, $\rho(n) := \mathcal{I}(\frac{3}{2}, 2n)/\mathcal{I}(\frac{3}{2}, 0) = (-\frac{3}{4})_n/(\frac{7}{4})_n$ is rational $(1, -\frac{3}{7}, -\frac{3}{77}, -\frac{1}{77}, -\frac{9}{1463}, \ldots)$, and since $\mathcal{I}(\frac{3}{2}, 0) = \frac{1}{3}\,\mathcal{I}(-\frac{1}{2}, 0) = \frac{\pi}{3}\gamma_{\mathrm{e}}$,

$$b^U_{mk} \;=\; \tfrac{1}{3}\gamma_{\mathrm{e}}\left[\rho(\tfrac{|m-k|}{2}) - \rho(\tfrac{m+k}{2}+1)\right] \;\in\; \mathbb{Q}\cdot\gamma_{\mathrm{e}}.$$

The *same single generator* γ_{e} serves both Chebyshev families — here without even the $\sqrt{2}$, since $\nu_n = \sqrt{2/\pi}$ uniformly for U. The diagonal is positive ($b^U_{00} = \frac{10}{21}\gamma_{\mathrm{e}}$, $b^U_{11} = \frac{80}{231}\gamma_{\mathrm{e}}$, $b^U_{22} = \frac{26}{77}\gamma_{\mathrm{e}}$, …) and every off-diagonal even entry is *negative* ($b^U_{02} = -\frac{10}{77}\gamma_{\mathrm{e}}$, $b^U_{13} = -\frac{32}{231}\gamma_{\mathrm{e}}$, $b^U_{04} = -\frac{2}{231}\gamma_{\mathrm{e}}$), verified by 40-digit quadrature.

6.2 Asymmetric case

For $(\alpha, \beta) = (\frac{1}{2}, -\frac{1}{2})$, $W^{3/2} = (1-x)^{3/4}(1+x)^{-3/4}$, which has a genuine algebraic singularity at $x = +1$. Here the orthonormal polynomials are Chebyshev polynomials of the fourth kind, $\widehat{P}_n = W_n/\sqrt{\pi}$ with $W_n(\cos\theta) = \sin((n+\frac{1}{2})\theta)/\sin(\theta/2)$, and the effective Wallis power is $p = 3(\alpha+\beta)/2 = 0$: the Gamma quotient collapses and the transcendental generator disappears entirely. The first row is

$$\sqrt{2}\,b_{0k} \;=\; 3,\; -\tfrac{3}{2},\; 1,\; -\tfrac{7}{8},\; \tfrac{3}{4},\; -\tfrac{11}{16},\; \ldots \;\in\; \mathbb{Q},$$

alternating in sign and decaying as $k^{-1/2}$ (exponent verified by quadrature, fitted slope -0.494 at $k = 64$). Unlike the symmetric cases the odd entries do *not* vanish: the broken $x \mapsto -x$ symmetry populates the full row.

7 The Sextic Test: $\alpha = \beta = -\frac{5}{9}$

The quartic arithmetic of the Chebyshev family — Pochhammer steps $(\frac{1}{4})_n/(\frac{3}{4})_n$ over the lemniscatic period $\Gamma(\frac{1}{4})^2$ — enters through the single Wallis power $p = -\frac{1}{2}$. Changing p should change the arithmetic, with thirds in place of quarters and the period of $y^2 = 1 - x^6$ in place of $y^2 = 1 - x^4$. This section confirms that prediction and identifies the general mechanism.

7.1 The Wallis ratio and the sextic generator

Proposition 7.1. *For $p > -1$ and $a = \frac{p+2}{2}$,*

$$\frac{\mathcal{I}(p, 2n)}{\mathcal{I}(p, 0)} \;=\; \frac{(1-a)_n}{(a)_n}.$$

Proof. By (1) the ratio is $(-1)^n\,\Gamma(a)^2/\big(\Gamma(a+n)\Gamma(a-n)\big) = (-1)^n/\big((a)_n\,(a-1)(a-2)\cdots(a-n)\big) = (1-a)_n/(a)_n$. □

For $p = -\frac{1}{2}$ this is the quartic ratio $r(n)$ of Proposition 3.1. Steps in thirds require $a \in \frac{1}{3} + \mathbb{Z}$. The matching integrand of a symmetric Jacobi weight has $p = 3\alpha + 1$, so the first such case is

$$\alpha = \beta = -\tfrac{5}{9}, \qquad p = -\tfrac{2}{3}, \qquad \frac{\mathcal{I}(-\frac{2}{3}, 2n)}{\mathcal{I}(-\frac{2}{3}, 0)} = \frac{(\frac{1}{3})_n}{(\frac{2}{3})_n}, \qquad \nu_0 := \mathcal{I}(-\tfrac{2}{3}, 0) = B(\tfrac{1}{6}, \tfrac{1}{2}) = \frac{2^{2/3}\pi\,\Gamma(\frac{1}{3})}{\Gamma(\frac{2}{3})^2}.$$

The generator ν_0 is a period of the curve $y^2 = 1 - x^6$, with complex multiplication by $\mathbb{Z}[\omega]$ — the sextic analogue of the lemniscatic $\Gamma(\frac{1}{4})^2$. (All exact statements in this section were verified against 40-digit quadrature of the defining integrals for $m, k \le 8$; see `jacobi_sextic.py` and its log.)

7.2 Two lattices

The orthogonal polynomials for $W = (1-x^2)^{-5/9}$ are Gegenbauer $C_n^{(\lambda)}$ with $\lambda = \alpha + \frac{1}{2} = -\frac{1}{18}$; unlike the Chebyshev cases they do not collapse to single cosines. The monic recurrence $\pi_{n+1} = x\pi_n - \beta_n \pi_{n-1}$ has

$$\beta_n = \frac{9n(9n-10)}{(18n-1)(18n-19)} \;=\; \frac{9}{17},\ \frac{144}{595},\ \frac{459}{1855},\ \frac{936}{3763},\ \cdots$$

so the polynomial side lives on *eighteenths* while the matching moments (Proposition 7.1) live on *thirds*. The matrix is a sandwich (eighteenths)^{T}(thirds)(eighteenths). The single quartic lattice of Chebyshev-T is the degeneration $\lambda = 0$, where $T_n \in \mathbb{Z}[x]$ makes the polynomial lattice invisible.

7.3 Exact data and integrality

Let PZ_n be the primitive integer polynomial proportional to π_n, so $\mathrm{PZ}_2 = 17x^2 - 9$, $\mathrm{PZ}_3 = 35x^3 - 27x$, $\mathrm{PZ}_4 = 1855x^4 - 1890x^2 + 243$; the leading coefficients are sliding windows of the primes $18j - 1$. Write $E_{mk} = \langle \mathrm{PZ}_m, \mathrm{PZ}_k \rangle_{W^{3/2}} / \nu_0$.

Table 2: Sextic matching matrix E_{mk} in units $\nu_0 = B(\frac{1}{6}, \frac{1}{2})$ (upper triangle, $0 \le m \le k \le 4$, primitive-$\mathbb{Z}$ polynomial frame). Entries with $m + k$ odd vanish identically.

$m\backslash k$	0	1	2	3	4
0	1	0	$\frac{15}{4}$	0	$\frac{621}{8}$
1		$\frac{3}{4}$	0	$\frac{27}{8}$	0
2			$\frac{1863}{40}$	0	$\frac{85455}{128}$
3				$\frac{5913}{128}$	0
4					$\frac{87779133}{2816}$

The denominators obey a sextic law. For all $m, k \le 8$ the odd part of $\mathrm{den}(E_{mk})$ divides $\prod_{j \le s+2}(6j-1)$ with $s = \frac{m+k}{2}$ — the sextic factorial family $5, 11, 17, 23, \ldots$ — and depends on s only. An s-dependent rescale therefore clears all denominators, exactly as $Q(s)$ does for Chebyshev-T; no pure row rescale exists.

7.4 The connection–Gram theorem

The structure behind both the quartic and the sextic case is classical. Set

$$\lambda = \alpha + \tfrac{1}{2}, \qquad \mu = \tfrac{3}{2}\alpha + \tfrac{1}{2}, \qquad \text{so} \qquad W^{3/2} = (1-x^2)^{\mu - \frac{1}{2}} \quad \text{and} \quad \lambda - \mu = -\tfrac{\alpha}{2}.$$

The weight $W^{3/2}$ is itself a Gegenbauer weight, of parameter μ.

Theorem 7.2 (Connection–Gram)**.** *Let $-\frac{2}{3} < \alpha$, $\alpha \ne -\frac{1}{2}$, and let $\pi_n = f_n C_n^{(\lambda)}$ with $f_n = n!/(2^n(\lambda)_n)$ be the monic Gegenbauer polynomials for $W = (1-x^2)^\alpha$. Then $\int \pi_m \pi_k W^{3/2} = 0$ for $m + k$ odd, and otherwise*

$$\int_{-1}^{1} \pi_m \pi_k \, W^{3/2}\, dx \;=\; f_m f_k \sum_{\substack{j \equiv m \pmod 2 \\ 0 \le j \le \min(m,k)}} A_{\frac{m-j}{2}}(m)\, A_{\frac{k-j}{2}}(k)\, h_j^\mu,$$

where

$$A_\ell(n) = \frac{(\lambda-\mu)_\ell\,(\lambda)_{n-\ell}\,(n-2\ell+\mu)}{\ell!\,(\mu)_{n-\ell+1}}, \qquad h_j^\mu = \int_{-1}^{1} \big(C_j^{(\mu)}\big)^2 (1-x^2)^{\mu-\frac{1}{2}} dx, \qquad \frac{h_j^\mu}{h_0^\mu} = \frac{(2\mu)_j\,\mu}{j!\,(j+\mu)}.$$

Proof. The classical connection formula for ultraspherical polynomials [4, §18.18(iv)] gives $C_n^{(\lambda)} = \sum_{\ell=0}^{\lfloor n/2 \rfloor} A_\ell(n)\, C_{n-2\ell}^{(\mu)}$. Substituting for both factors and using the orthogonality of the $C_j^{(\mu)}$ with respect to $(1-x^2)^{\mu-1/2}$ collapses the double sum to the matched indices $j = m-2\ell = k-2\ell'$. □

The matching matrix is thus the *Gram matrix of the connection coefficients* $C^{(\lambda)} \to C^{(\mu)}$ in the μ-norm. Every Pochhammer that appears steps by λ, μ, 2μ, or $\lambda - \mu = -\frac{\alpha}{2}$; for $\alpha = -\frac{5}{9}$ these are the eighteenths and thirds of the two lattices.

Corollary 7.3 (Row 0)**.** *With* $h_0^\mu = \mathcal{I}(3\alpha+1, 0)$*,*

$$\frac{\langle \pi_0, \pi_{2n} \rangle_{W^{3/2}}}{h_0^\mu} \;=\; \frac{(2n)!\,(\lambda-\mu)_n\,(\lambda)_n\,\mu}{2^{2n}\,(\lambda)_{2n}\,n!\,(\mu)_{n+1}}.$$

At $\alpha = -\frac{5}{9}$ the factor $(\lambda-\mu)_n = (\frac{5}{18})_n$ explains the primes $5, 23, 41, 59 \equiv 5 \pmod{18}$ observed in the row-0 numerators of Table 2.

7.5 Chebyshev as rational specialisation

The excluded value $\alpha = -\frac{1}{2}$ is where $\lambda = 0$ and $C_n^{(\lambda)}$ vanishes. The monic frame removes the singularity, since the pole of $f_n = n!/(2^n(\lambda)_n)$ cancels the zeros of $(\lambda)_{n-\ell}$ in $A_\ell(n)$.

Corollary 7.4. *Each normalised entry* $\langle \pi_m, \pi_k \rangle_{W^{3/2}} / \mathcal{I}(3\alpha+1, 0)$*, and each side of Theorem 7.2, is a rational function of* α*; the theorem therefore extends to* $\alpha = -\frac{1}{2}$ *by evaluation, where it reproduces Proposition 3.1. In the integer form, for* $m, k \geq 1$ *with* $m+k$ *even,*

$$Q(s)\, 2^{m+k-2}\, \frac{\langle \pi_m, \pi_k \rangle_{W^{3/2}}}{\mathcal{I}(-\frac{1}{2}, 0)}\bigg|_{\alpha=-\frac{1}{2}} \;=\; \frac{1}{2}\left[\frac{P(d)\,Q(s)}{Q(d)} + P(s)\right],$$

with $P(n) = 1 \cdot 5 \cdots (4n-3)$*,* $Q(n) = 3 \cdot 7 \cdots (4n-1)$*,* $s = \frac{m+k}{2}$*,* $d = \frac{|m-k|}{2}$*.*

Proof. Moment ratios are rational in α by Proposition 7.1, hence so are the monic Gram–Schmidt data and the normalised entries; the right-hand side of Theorem 7.2 is rational in α by inspection. Two rational functions agreeing on $(-\frac{2}{3}, \infty) \setminus \{-\frac{1}{2}\}$ agree at $-\frac{1}{2}$. The evaluation is Proposition 3.1 in monic units ($T_n = 2^{n-1}\widetilde{T}_n$). □

The two-term Chebyshev closed form thus acquires a second, structural derivation. At $\lambda = 0$ the degenerate connection coefficients are those of T_n in the basis $C^{(-1/4)}$, and the Gram sum telescopes to the two Wallis terms $r(d) + r(s)$. The specialisation was checked symbolically for all $m, k \leq 8$ (`jacobi_lambda0_limit.py`); the degenerate coefficients reconstruct T_n exactly.

8 Generating Function and Combinatorics

8.1 Generating function for Chebyshev-T

The double generating function $G(w,u) = \sum_{m,k\geq 0} b_{mk}^T w^m u^k$ can be evaluated exactly. Using $T_m(x) = \frac{1}{2}(z^m + z^{-m})$ with $z = e^{i\theta}$ and the geometric series $\sum_{m\geq 0}(we^{im\theta} + we^{-im\theta}) = \mathrm{Re}\frac{2w}{1-we^{i\theta}}$, one obtains

$$G(w,u) = \frac{\nu_0^2}{\pi}\int_0^\pi \frac{\sin^{-1/2}\theta}{\left(1 - 2w\cos\theta + w^2\right)\left(1 - 2u\cos\theta + u^2\right)}\, d\theta \quad + \text{ boundary corrections},$$

which reduces to a complete elliptic integral of the third kind $\Pi(n,k) = \int_0^{\pi/2}(1 - n\sin^2\phi)^{-1}(1 - k^2\sin^2\phi)^{-1/2} d\phi$ for $|w|, |u| < 1$.

8.2 OEIS content of rational coefficients

The diagonal rationals are $b^T_{nn}/\gamma_e = 1 + r(n)$:

$$1,\ \tfrac{4}{3},\ \tfrac{26}{21},\ \tfrac{92}{77},\ \tfrac{90}{77},\ \tfrac{1684}{1463},\ \ldots$$

The denominators $1, 3, 21, 77, 77, 1463, \ldots$ are (after cancellation) the **quartic factorials** $Q(n) = 3 \cdot 7 \cdot 11 \cdots (4n-1)$ (OEIS A008545). The first row gives the pure quartic-factorial ratios

$$\frac{b^T_{0,2n}}{\sqrt{2}\,\gamma_e} \;=\; r(n) \;=\; \frac{P(n)}{Q(n)} \;=\; \frac{1 \cdot 5 \cdot 9 \cdots (4n-3)}{3 \cdot 7 \cdot 11 \cdots (4n-1)} \;=\; 1,\ \tfrac{1}{3},\ \tfrac{5}{21},\ \tfrac{15}{77},\ \tfrac{13}{77},\ \ldots$$

with numerators $P(n) = 1 \cdot 5 \cdots (4n-3)$ in OEIS A007696. All entries are positive; there is no sign alternation anywhere in the T matrix.

9 Zeros: Crowding Toward the Bernstein Ellipse $\rho \to 1^+$

9.1 Setup

The truncated approximant is $A_N(x) = \sum_{k=0}^{N} b_{0k} \widehat{P}^T_k(x)$, a polynomial of degree N. Its N zeros in $\mathbb{C}$ would split into *genuine* zeros (tracking zeros of the target function) and *spurious* zeros; here the target $\Phi_0 = W^{1/2}$ is zero-free, so all N zeros are spurious.

9.2 Bernstein ellipse background

For a function analytic in the Bernstein ellipse $\mathcal{E}_\rho = \{z = (\rho e^{i\theta} + \rho^{-1} e^{-i\theta})/2 : \theta \in [0, 2\pi]\}$ (semi-axes $a = (\rho + \rho^{-1})/2$, $b = (\rho - \rho^{-1})/2$), the Chebyshev coefficients decay geometrically as $c_k \sim \rho^{-k}$. If instead $c_k \sim k^{-s}$ (power law), then the "effective" Bernstein parameter ρ_N satisfying $\rho_N^{-N} \sim N^{-s}$ gives $\rho_N = \exp(s \log N/N) \to 1^+$.

Theorem 9.1 (Bernstein crowding)**.** *For Chebyshev-T, the spurious zeros of A_N accumulate on $[-1, 1]$ as $N \to \infty$: all spurious zeros satisfy $|z| \le 1 + O(\log N/N)$. The Bernstein parameter satisfies $\rho_N \to 1^+$ at rate $\log N/N$.*

This contrasts sharply with the Hermite case (geometric decay, $\rho_N \to \sqrt{3} > 1$) and parallels the MP case (power-law decay, $\rho_N \to 1^+$ at a slower or faster rate depending on the exponent).

9.3 Numerical evidence

Figure 1 shows the zeros of A^T_N and A^U_N for $N = 10, 20, 40$. Since the target $\Phi_0 = W^{1/2}$ is zero-free on $(-1, 1)$, *every* zero is spurious, and indeed none are real: for Chebyshev-T (left panel) all N zeros are complex, with moduli filling out [0.1, 0.994] at $N = 40$ and accumulating toward the segment as N grows. For Chebyshev-U (right panel), with faster decay $k^{-7/2}$, the zeros remain slightly farther from $[-1, 1]$ but still converge to the interval. (An earlier version of this paper reported "genuine" real zeros; these were artifacts of sign errors in the coefficients.)

10 The Weight Triptych

Table 3 summarises the three families studied.
The key contrasts:

- **Decay**: determined by the furthest analytic singularity. Entire weights give geometric decay; pole singularities give power-law $k^{-\alpha}$ (exponent from the distance to the pole); endpoint singularities give power-law with exponent derived from the endpoint exponent via Stirling — shifted by one when two Wallis terms cancel at leading order (the U case).

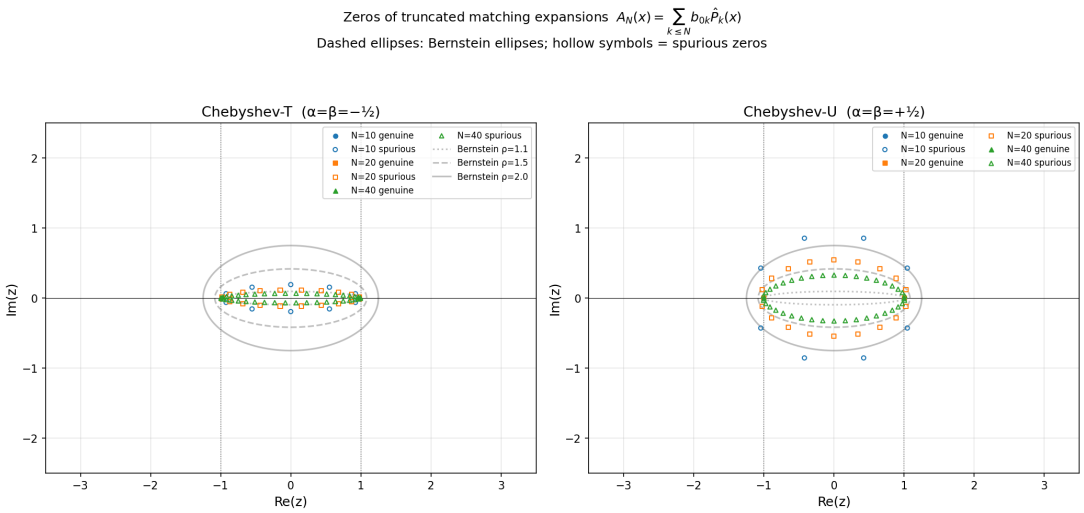


Figure 1: Zeros of truncated matching expansions $A_N(x) = \sum_{k \le N} b_{0k} \widehat{P}_k(x)$ for $N = 10, 20, 40$. **Left**: Chebyshev-T ($\alpha = \beta = -\frac{1}{2}$), decay $k^{-1/2}$. **Right**: Chebyshev-U ($\alpha = \beta = +\frac{1}{2}$), decay $k^{-7/2}$. All zeros are spurious (the target $\sqrt{W}$ is zero-free); none are real. Gray ellipses: Bernstein ellipses $\mathcal{E}_\rho$ for $\rho = 1.1, 1.5, 2.0$. As $N \to \infty$, spurious zeros crowd toward $\mathcal{E}_1 = [-1, 1]$.

Table 3: Triptych comparison: three weight families.

Weight	Supp.	Analytic structure	Decay b_{0k}	Generator	Spur. zeros
Hermite e^{-x^2}	$\mathbb{R}$	Entire	$3^{-k/2}$ geometric	$\mathbb{Z}$	$\rho = \sqrt{3}$
Meixner–Poll.	$\mathbb{R}$	Poles at $\pm i/4$	$k^{-3/4}$ power law	$\mathbb{R}$ (open)	Bump–Ng
Jacobi $W_{\alpha\beta}$	$[-1, 1]$	Branch pts at ± 1	$k^{-1/2}$ (T), $k^{-7/2}$ (U)	$\mathbb{Q}(\sqrt{2}) \cdot \Gamma(\frac{1}{4})^2$	$\rho_N \to 1^+$

- **Generator ring**: Hermite is integer ($\mathbb{Z}$), reflecting the combinatorial structure of Gaussian integration. Jacobi is $\mathbb{Q}(\sqrt{2}) \cdot \Gamma(\frac{1}{4})^2/\pi^{3/2}$, reflecting the lemniscate constant $\varpi = \Gamma(\frac{1}{4})^2/(4\sqrt{2\pi})$. The MP case remains empirically $\mathbb{R}$; identifying its algebraic structure is open.
- **Zero accumulation**: Geometric decay $\Rightarrow$ fixed Bernstein parameter; power-law decay $\Rightarrow$ crowding $\rho_N \to 1^+$ at rate $(\log N)/N$ (slower for larger exponent).

11 Discussion

11.1 Lemniscate constant and arithmetic

The generator $\Gamma(\frac{1}{4})^2$ is related to the **lemniscate constant** $\varpi = \Gamma(\frac{1}{4})^2/(4\sqrt{2\pi}) \approx 2.6221$ and to the arithmetic-geometric mean $\mathrm{agm}(1, \sqrt{2}) = \pi\sqrt{2}/\Gamma(\frac{1}{4})^2$. The appearance of $\Gamma(\frac{1}{4})^2$ in matching-matrix arithmetic mirrors its role in elliptic curves with complex multiplication by $\mathbb{Z}[i]$ (the elliptic curve $y^2 = 1 - x^4$). The sextic case of §7 supplies the companion example, with $B(\frac{1}{6}, \frac{1}{2})$, the curve $y^2 = 1 - x^6$, and CM by $\mathbb{Z}[\omega]$; matching arithmetic appears to be *period arithmetic* of the weight.

11.2 Open problems

1. **General decay law**: Theorem 4.1 settles the Chebyshev cases exactly ($C_T = 2/\sqrt{\pi}$, $C_U = 15/(2\sqrt{2\pi})$). For general (α, β), when does leading-order cancellation (as in the U case) shift the naive exponent $-(3(\alpha+\beta)/2+1)$?

2. **Integrality**: For which normalisations of the Jacobi weight does the matching matrix have integer (or half-integer) entries? (Solved for the Chebyshev family in the companion synthesis paper: rescaling by $Q(s)$ makes T integral, by $3Q^+(s+1)$ with $Q^+(n) = 7 \cdot 11 \cdots (4n+3)$ makes U integral, and the asymmetric case is dyadic — $2^k\sqrt{2}\, b_{0k}$ gives twice central binomials for even k and $-(4n+3)\,\mathrm{Cat}(n)$ for odd $k = 2n+1$.)

3. **Exact Bernstein parameter**: Does $\rho_N = (1 + s \log N/N)(1 + O(1/N))$ with $s =$ (exponent)? Numerical evidence suggests yes.

4. **MP generator ring**: Is the Meixner–Pollaczek matching generator algebraically related to π or $\Gamma(\frac{1}{4})$?

5. **General (α, β) and CM classification**: The connection–Gram mechanism of §7 extends verbatim to asymmetric Jacobi weights via the Jacobi connection coefficients. Which exponents produce CM periods, and what replaces the two-lattice structure when $\alpha \neq \beta$?

11.3 Code availability

All computations used `mpmath` at 50 decimal digits. The scripts `jacobi_matching.py` and `jacobi_zeros.py` are available in `github.com/RaggedR/category`, together with the exact `sympy` verifications of §7 (the `jacobi_sextic*` and `jacobi_lambda0_limit` scripts).

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